Choice-Free Reverse Mathematics.

From Barrow's Geometry to the Recursive Comprehension Axiom RCA_0

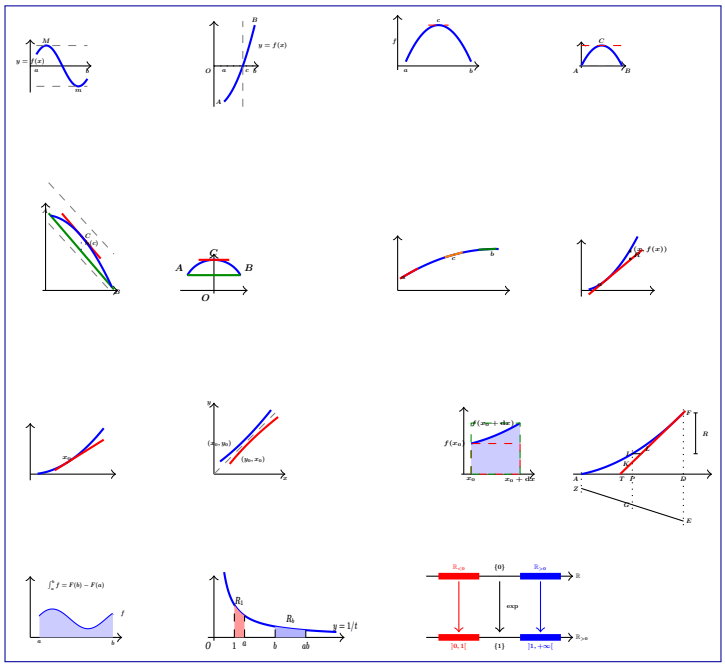
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Abstract

Part 1 develops classical single-variable real analysis in Barrow's geometric language of areas, ordinates, and subtangents. Part 2 classifies each theorem in reverse mathematics. MVT and Rolle are RCA_0 via Hardin–Velleman; the mean-value theorem for integrals, the inverse-function theorem (Yokoyama's strong-C^1 form), and the ln/exp chain are RCA_0; IVT, EVT, Darboux, and the weighted MVT are WKL_0 as proved here. All C^k hypotheses are read as strong C^k-coding. No entry uses Cauchy completeness or any form of Choice.



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Introduction

Barrow's *Geometrical Lectures* (1670) give the first calculus textbook and the first geometric proof of the Fundamental Theorem of Calculus; see [1, 2].

Part 1 develops the classical theorems of single-variable real analysis — EVT, IVT, Fermat, Rolle, MVT and its refinements, convexity, inverse function, the Fundamental Theorems of Calculus, and the ln/exp addition and image theorems — in Barrow's geometric idiom, with figures and geometric proofs.

Part 2 classifies each theorem in reverse mathematics. Every axiom, postulate, definition and statement of Part 1 reappears as a tagged entry with a dependency table and an exact strength. Classification: all entries are RCA_0 except IVT, EVT, Darboux, and the weighted MVT, which are WKL_0. No entry uses Cauchy completeness or any form of Choice [7, 14]. Coding conventions are in §5.

Part 1

A Geometric Development of Classical Real Analysis

Following Barrow [1, Lecture X, §11], we state and prove the classical theorems in Barrow's geometric idiom. See [2] for commentary.

1. Axioms, definitions and geometric conventions

Let $[a,\,b] \subseteq \mathbb{R}$ and let $f\colon [a,\,b] \to \mathbb{R}$ be continuous. The *area function* is

$$A_a^x := \text{(signed) area of } \{(t,\,y) : a \le t \le x,\ \ 0 \le y \le f(t)\}, \tag{1}$$

with Barrow's orientation

$$A_x^x = 0, \qquad A_x^{x+\mathrm{d}x} = -A_{x+\mathrm{d}x}^{x} \quad (\mathrm{d}x < 0). \tag{2}$$

The curve $u \mapsto A_a^u$ is Barrow's AIF; $f(x_0)$ is his DE; the unit is his R.

Differentiability. f is differentiable at x if

$$\lim_{h \to 0} (f(x+h) - f(x))/h = f'(x) \tag{3}$$

exists.

Tangency (little-o form). G is differentiable at x_0 with tangent $L(u) = m(u - x_0) + G(x_0)$ iff

$$G(x_0 + \Delta x) = G(x_0) + m\,\Delta x + \Delta x \cdot o(\Delta x) \quad (\Delta x \to 0), \tag{4}$$

in which case $G'(x_0) = m$; cf. [2, Theorem 2].

Lin-Integ.

$$\forall u,\, v,\, w \in [a,\, b] :\ A_u^w = A_u^v + A_v^w. \tag{5}$$

Int-Mono.

$$f \le g \text{ on } [a,\, b] \Rightarrow \int_a^b f \le \int_a^b g. \tag{6}$$

Similarity. For $b > 0$,

$$\int_b^{ab} h(t)\,\mathrm{d}t = \int_1^a h(bu)\,b\,\mathrm{d}u. \tag{7}$$

Inv-Cont.

$$t \mapsto 1/t \in C^0(\mathbb{R}_{>0}). \tag{8}$$

Image-Union.

$$f(A \cup B) = f(A) \cup f(B). \tag{9}$$

Product rule.

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x). \tag{10}$$

Chain rule.

$$(f \circ g)'(x) = f'(g(x))\,g'(x). \tag{11}$$

Sign-Law.

$$\forall u > 0 :\ (\ln u < 0 \iff u < 1) \wedge (\ln u = 0 \iff u = 1) \wedge (\ln u > 0 \iff u > 1). \tag{12}$$

Ln-Bij. $\ln : \mathbb{R}_{>0} \to \mathbb{R}$ is bijective; $\exp := \ln^{-1} : \mathbb{R} \to \mathbb{R}_{>0}$, i.e.

$$\exp := \ln^{-1} : \mathbb{R} \to \mathbb{R}_{>0}. \tag{13}$$

Barrow-tangency. Let G be differentiable at x_0, and let $L(u) = m(u - x_0) + G(x_0)$ be the line through $(x_0,\, G(x_0))$ of slope m. If $L \le G$ (or $L \ge G$) on a punctured neighbourhood of x_0, then[1]

$$m = G'(x_0). \tag{14}$$

2. Statements of the classical theorems

2.1. Topological theorems

Theorem 1 (Extreme Value Theorem).

If $f : [a,\, b] \to \mathbb{R}$ is continuous, then

$$\exists x_{\max},\, x_{\min} \in [a,\, b],\ \ \forall x \in [a,\, b] :\ f(x_{\min}) \le f(x) \le f(x_{\max}). \tag{15}$$

[1] The two-sided hypothesis strengthens Barrow's one-sided criterion, which fails at inflections [2]. Every use in Part 1 is in a convex or concave setting, where the two-sided hypothesis holds.

Theorem 2 (Intermediate Value Theorem).
If $f : [a,\, b] \to \mathbb{R}$ is continuous and $f(a)f(b) < 0$, then

$$\exists c \in (a,\, b) :\ \ f(c) = 0. \tag{16}$$

2.2. Differentiation theorems

Theorem 3 (Fermat).
If f has a local extremum at $c \in (a,\, b)$ and is differentiable at c, then

$$f'(c) = 0. \tag{17}$$

Theorem 4 (Rolle).
If f is continuous on $[a,\, b]$, differentiable on $(a,\, b)$, and $f(a) = f(b)$, then

$$\exists c \in (a,\, b) :\ f'(c) = 0. \tag{18}$$

Theorem 5 (Mean-Value Theorem).
If f is continuous on $[a,\, b]$ and differentiable on $(a,\, b)$, then

$$\exists c \in (a,\, b) :\ \ f'(c) = \frac{f(b) - f(a)}{b - a}. \tag{19}$$

Geometrically, $d_c \parallel d_{a,\,b}$.

Theorem 6 (Constant-function).
If f is continuous on $[a,\, b]$, differentiable on $(a,\, b)$, and $f' = 0$ on $(a,\, b)$, then

$$f(x) = f(a) \quad \text{for all } x \in [a,\, b]. \tag{20}$$

Theorem 7 (Monotonicity).
If f is continuous on $[a,\, b]$, differentiable on $(a,\, b)$, and $f' \geq 0$ (resp. $f' > 0$), then

$$x < y \Rightarrow f(x) \leq f(y) \quad (\text{resp. } f(x) < f(y)) \tag{21}$$

for all $x,\, y \in [a,\, b]$.

Theorem 8 (Cauchy MVT).
If $f,\, g$ are continuous on $[a,\, b]$, differentiable on $(a,\, b)$, and $g' \neq 0$ on $(a,\, b)$, then

$$\exists c \in (a,\, b) :\ \ \frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}. \tag{22}$$

Theorem 9 (Darboux).
If f is differentiable on $[a,\, b]$ and $f'(a) < \lambda < f'(b)$, then

$$\exists c \in (a,\, b) :\ \ f'(c) = \lambda. \tag{23}$$

Theorem 10 (L'Hôpital).
Let f, g be differentiable near a, with $f(a) = g(a) = 0, g' \neq 0$ near a, and

$$\lim_{x \to a} f'(x)/g'(x) = L.$$

Then

$$\lim_{x \to a} f(x)/g(x) = L. \tag{24}$$

Theorem 11 (Taylor, Lagrange remainder).
If f is $(n+1)$-times differentiable on an interval containing a and x, then

$$\exists c \text{ between } a,\ x:\ f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}. \tag{25}$$

Theorem 12 (Convexity).
If f is twice differentiable on $[a,\ b]$ and $f'' \geq 0$ on $(a,\ b)$, then

$$f(x) \geq f(x_0) + f'(x_0)(x - x_0) \quad \text{for all } x,\ x_0 \in [a,\ b]. \tag{26}$$

Theorem 13 (Inverse function).
If $f : I \to \mathbb{R}$ is continuous and strictly monotone on I, x_0 lies in the interior I° of I, and f is differentiable at x_0 with $f'(x_0) \neq 0$, then f^{-1} is differentiable at $y_0 = f(x_0)$ with

$$(f^{-1})'(y_0) = 1/f'(x_0). \tag{27}$$

2.3. Integration theorems

Lemma 14 (Barrow's area inequality).
If f is continuous and increasing on $[a,\ b]$ and $x_0,\ x_0 + \mathrm{d}x \in [a,\ b], \mathrm{d}x \neq 0$, then

$$f(x_0)\,\mathrm{d}x \leq A_{x_0}^{x_0+\mathrm{d}x} \leq f(x_0 + \mathrm{d}x)\,\mathrm{d}x, \tag{28}$$

and both inequalities are strict when f is strictly increasing on the interval with endpoints x_0 and $x_0 + \mathrm{d}x$.

Lemma 15 (MVT for integrals).
If f is continuous on $[a,\ b]$ and $x_0,\ x_0 + \mathrm{d}x \in [a,\ b], \mathrm{d}x \neq 0$, then

$$\exists \theta \text{ between } x_0,\ x_0 + \mathrm{d}x:\ A_{x_0}^{x_0+\mathrm{d}x} = f(\theta)\,\mathrm{d}x. \tag{29}$$

Theorem 16 (First FTC; Barrow).
If f is continuous on $[a,\ b]$, then

$$\frac{d}{dx}\int_a^x f = f(x) \tag{30}$$

for every $x \in (a,\ b)$, with one-sided versions at a, b.

Theorem 17 (Second FTC).
If F is differentiable on $[a,\,b]$ and $F' = f$ is continuous, then

$$\int_a^b f = F(b) - F(a). \tag{31}$$

Theorem 18 (Integration by parts).
If $f,\, g \in C^1([a,\,b])$, then

$$\int_a^b fg' = [fg]_a^b - \int_a^b f'g. \tag{32}$$

Theorem 19 (Change of variables).
If $g \in C^1([a,\,b])$ and f is continuous on $g([a,\,b])$, then

$$\int_{g(a)}^{g(b)} f = \int_a^b f(g)\, g'. \tag{33}$$

Theorem 20 (Weighted MVT).
If f is continuous on $[a,\,b], w \geq 0$ is Riemann integrable, and $\int_a^b w > 0$, then

$$\exists c \in [a,\,b] : \int_a^b fw = f(c) \int_a^b w. \tag{34}$$

2.4. The logarithm and the exponential

Theorem 21 (Log-Property).
With $\ln(x) := \int_1^x \mathrm{d}t/t$ $(x > 0)$,

$$\forall a,\, b > 0 : \ \ln(a) + \ln(b) = \ln(ab). \tag{35}$$

Theorem 22 (Exp-Property).
With $\exp := \ln^{-1}$,

$$\forall u,\, v \in \mathbb{R} : \ \exp(u)\exp(v) = \exp(u+v). \tag{36}$$

Theorem 23 (Exp-Image).

$$\exp(\mathbb{R}) = \mathbb{R}_{>0}, \qquad \ln(\mathbb{R}_{>0}) = \mathbb{R}. \tag{37}$$

3. Geometrical proofs

3.1. Topological theorems

Proof of Theorem 1. $S = f([a,\,b])$ is nonempty and bounded; let $M = \sup S, m = \inf S$. Were M unattained, $1/(M - f)$ would be continuous and unbounded on $[a,\,b]$; contradiction. So $M = f(x_{\max})$; symmetrically $m = f(x_{\min})$, giving Equation 15. □

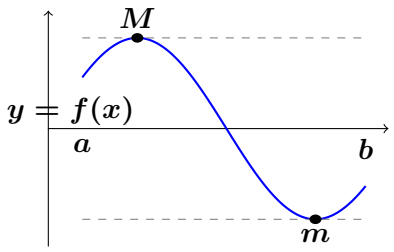


Figure 1. EVT.

Proof of Theorem 2. Assume $f(a) < 0 < f(b)$ and let $c = \sup\{x \in [a, b] : f(x) < 0\}$. By continuity $f(c) \leq 0$; if $f(c) < 0$ then $f < 0$ on a neighbourhood of c intersected with $[a, b]$, contradicting the definition of c as an upper bound; hence Equation 16 holds. □

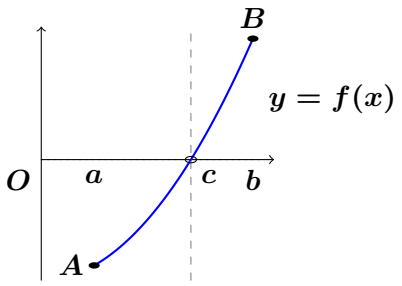


Figure 2. IVT.

3.2. Differentiation theorems

Proof of Theorem 3. If c is a local max: $(f(c+h)-f(c))/h \leq 0$ for $h > 0^+$ gives $f'(c) \leq 0$; for $h < 0^-, \geq 0$ gives $f'(c) \geq 0$. So $f'(c) = 0$. □

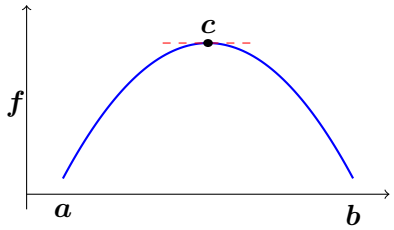


Figure 3. Fermat.

Proof of Theorem 4. By EVT, f attains M, m. If $M = m$, f constant. Otherwise, $f(a) = f(b)$ forces an interior extremum c; Fermat gives $f'(c) = 0$. □

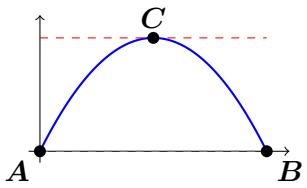


Figure 4. Rolle.

Proof of Theorem 5 (Barrow's sliding tangent). Let L be the chord line, $g = f - L$. Then $g(a) = g(b) = 0$; by EVT, g attains M, m. *Case* $M > 0$: maximum at $c \in (a, b)$; the translate $L + M$ touches f at C and lies above; by concave Barrow-tangency, slope $= f'(c) = L'(c) = (f(b) - f(a))/(b - a)$. *Case* $M = 0$: if $f = L$ identically, trivial; otherwise $m < 0$ with minimum at interior c and the same argument via convex Barrow-tangency. So $d_c \parallel d_{a,b}$. □

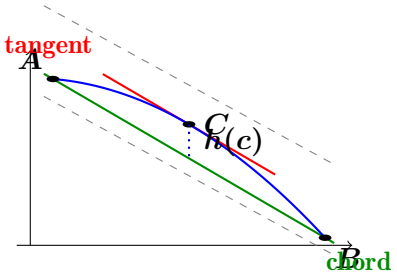


Figure 5. MVT sliding tangent.

Proof of Theorem 6. MVT on $[a, x]$: $f(x) - f(a) = f'(c)(x - a) = 0$. □

Proof of Theorem 7. MVT on $[x_1, x_2]$: $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$; sign of $f'(c)$ gives the result. □

Proof of Theorem 8. $h := (f(b) - f(a))g - (g(b) - g(a))f$ satisfies $h(a) = h(b)$, so Rolle gives c with $(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$; g' and $g(b) - g(a)$ are nonzero, so Equation 22. □

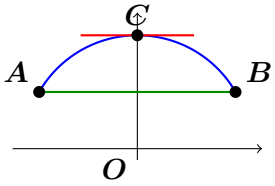


Figure 6. Cauchy MVT.

Proof of Theorem 9. $g = f - \lambda x$ has $g'(a) < 0 < g'(b)$. Small h gives $g(a + h) < g(a)$ and $g(b - h) < g(b)$; by EVT the minimum of g is interior, at c. Fermat: $g'(c) = 0$, i.e. $f'(c) = \lambda$. □

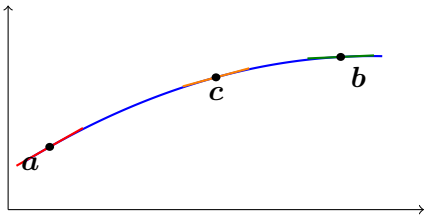


Figure 7. Darboux.

Proof of Theorem 10. Cauchy MVT on $[a, x]$: $f(x)/g(x) = f'(c_x)/g'(c_x)$; $c_x \to a$ as $x \to a^+$. Symmetric for $x \to a^-$. So the limit is L. □

Proof of Theorem 11. $P := \sum f^{(k)}(a)(t-a)^k/k!$; choose M with $f(x) = P(x) + M(x-a)^{n+1}$; $h = f - P - M(t-a)^{n+1}$ has $h^{(j)}(a) = 0, h(x) = 0$. Rolle $n+1$ times: $h^{(n+1)}(c) = 0$, so $M = f^{(n+1)}(c)/(n+1)!$. □

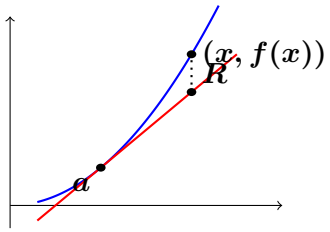


Figure 8. Taylor ($n = 1$).

Proof of Theorem 12. $g = f - f(x_0) - f'(x_0)(x - x_0)$ has $g'' \geq 0$; g' nondecreasing and $g'(x_0) = 0$, so $g' \leq 0$ before and ≥ 0 after x_0; with $g(x_0) = 0, g \geq 0$. Hence Equation 26. □

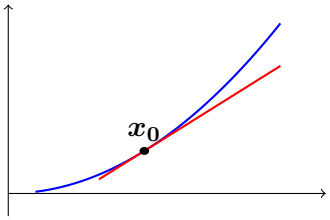


Figure 9. Convexity.

Proof of Theorem 13. Let $c := f'(x_0) \neq 0$. Since $x_0 \in I^\circ$, choose $\rho > 0$ with $(x_0 - \rho, x_0 + \rho) \subseteq I$. *Local surjectivity.* Assume without loss of generality that f is strictly increasing on I (the strictly decreasing case is symmetric). Choose $x_0^-, x_0^+ \in (x_0 - \rho, x_0 + \rho)$ with $x_0^- < x_0 < x_0^+$; then $f(x_0^-) < y_0 < f(x_0^+)$. By the intermediate value theorem applied to the continuous function f on the closed subinterval $[x_0^-, x_0^+] \subseteq I$, every $y \in (f(x_0^-), f(x_0^+))$ is attained at some $x \in (x_0^-, x_0^+) \subseteq I$. Hence $(f(x_0^-), f(x_0^+)) \subseteq f(I)$ is an open neighbourhood of y_0, and f^{-1} is defined on it.

Continuity of f^{-1} at y_0. Differentiability of f at x_0 gives, for some $\delta_0 \in (0,\ \rho)$ and all x with $|x - x_0| < \delta_0$,

$$|f(x) - f(x_0) - c(x - x_0)| < \tfrac{|c|}{2}|x - x_0|,$$

whence by the reverse triangle inequality $|f(x)-f(x_0)| \geq \tfrac{|c|}{2}|x-x_0|$ on the same neighbourhood. Given $\varepsilon > 0$, set $\varepsilon' := \min(\varepsilon,\ \delta_0)$ and

$$\eta := \min\Big(\tfrac{|c|}{2}\varepsilon',\ \ y_0 - f(x_0^-),\ \ f(x_0^+) - y_0\Big) > 0.$$

Suppose $|y - y_0| < \eta$. Since $y \in (f(x_0^-),\ f(x_0^+)) \subseteq f(I)$, the value $x := f^{-1}(y) \in (x_0 - \rho,\ x_0 + \rho) \subseteq I$ is defined. In the case $|x - x_0| < \delta_0$, the bound above gives $|y - y_0| \geq \tfrac{|c|}{2}|x - x_0|$, so $|x - x_0| < \varepsilon' \leq \varepsilon$. In the case $|x - x_0| \geq \delta_0$, let $x_1 := x_0 + \operatorname{sgn}(x - x_0)\,\delta_0 \in (x_0 - \rho,\ x_0 + \rho) \subseteq I$ be the point of I at distance exactly δ_0 from x_0 on the side of x. By strict monotonicity of f, $|y-y_0| \geq |f(x_1)-f(x_0)|$. Passing the bound $|f(u)-f(x_0)| \geq \tfrac{|c|}{2}|u-x_0|$, valid for $|u - x_0| < \delta_0$, to the limit $u \to x_1$ from inside $(x_0,\ x_1)$ (using continuity of f at x_1) yields $|f(x_1) - f(x_0)| \geq \tfrac{|c|}{2}\delta_0 \geq \tfrac{|c|}{2}\varepsilon' \geq \eta$, contradicting $|y - y_0| < \eta$. Hence $|y - y_0| < \eta \Rightarrow |f^{-1}(y) - x_0| < \varepsilon$, so f^{-1} is continuous at y_0.

Derivative of f^{-1} at y_0. Changing variable $y = f(x)$, i.e. $x = f^{-1}(y) \to x_0$ as $y \to y_0$ (continuity of f^{-1} at y_0, just established, justifies the substitution),

$$(f^{-1})'(y_0) = \lim_{y \to y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \lim_{x \to x_0} \frac{x - x_0}{f(x) - f(x_0)} = \frac{1}{f'(x_0)},$$

which is Equation 27. □

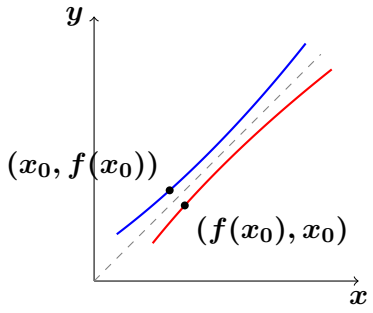


Figure 10. Inverse.

3.3. Integration theorems

Proof of Theorem 14. For $\mathrm{d}x > 0$, $f(x_0) \leq f \leq f(x_0 + \mathrm{d}x)$ on the interval; inscribed/circumscribed rectangles bound A. For $\mathrm{d}x < 0$, use Equation 2. □

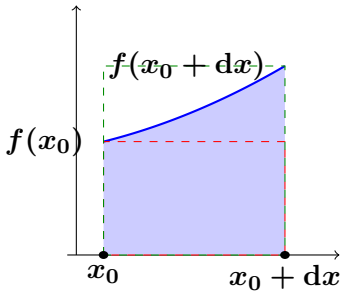


Figure 11. Barrow's area inequality.

Proof of Theorem 15. $G(t) = A_{x_0}^t$ is differentiable with $G' = f$; MVT on the interval gives θ with $A_{x_0}^{x_0+\mathrm{d}x} = f(\theta)\mathrm{d}x$. □

Proof of Theorem 16. Fix $x_0 \in (a, b)$ and $\varepsilon > 0$. Let $h \neq 0$ with $x_0 + h \in [a, b]$, and let I_h be the closed interval with endpoints x_0 and $x_0 + h$. Since f is continuous at x_0, for $|h|$ sufficiently small there are constants m_ε, M_ε with $m_\varepsilon \leq f(u) \leq M_\varepsilon$ on I_h, $|m_\varepsilon - f(x_0)| < \varepsilon$, $|M_\varepsilon - f(x_0)| < \varepsilon$. By integral monotonicity applied on I_h to the constant functions m_ε and M_ε, for $h > 0$ one has $m_\varepsilon h \leq A_{x_0}^{x_0+h} \leq M_\varepsilon h$; for $h < 0$ the same inequality holds with both directions reversed (the oriented convention Equation 2 changes the sign of both $A_{x_0}^{x_0+h}$ and h). In either case, dividing by h preserves the direction of the inequality (both sides change sign together with h), giving

$$m_\varepsilon \;\leq\; \frac{A_a^{x_0+h} - A_a^{x_0}}{h} \;\leq\; M_\varepsilon,$$

where the numerator uses additivity Equation 5. Since $\varepsilon > 0$ was arbitrary and m_ε, $M_\varepsilon \to f(x_0)$ as $\varepsilon \to 0$, the squeeze yields $\frac{d}{dx} A_a^x|_{x_0} = f(x_0)$; since $x_0 \in (a, b)$ was arbitrary, Equation 30 holds on (a, b), and the one-sided cases at $x = a$, b follow by one-sided limits. □

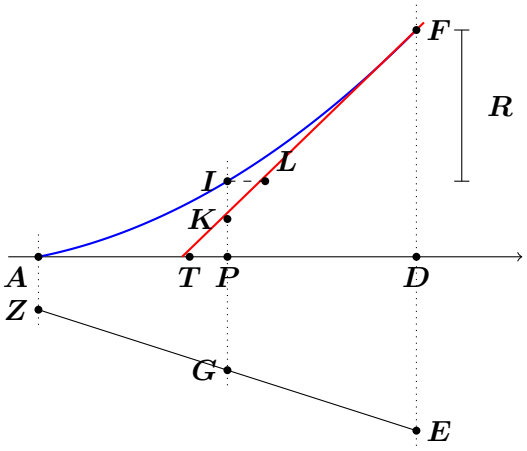


Figure 12. Barrow's Fig. 109.

Proof of Theorem 17. $G(x) = \int_a^x f$ has $G' = f$; so $(F - G)' = 0$ and $F - G$ is constant; $F(b) - F(a) = \int_a^b f$. □

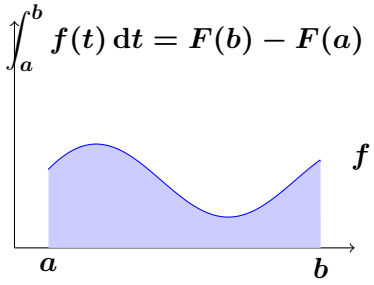


Figure 13. Second FTC.

Proof of Theorem 18. $(fg)' = f'g + fg'$; integrate with Second FTC. □

Proof of Theorem 19. Let $F' = f$. Chain rule: $(F \circ g)' = f(g)g'$; Second FTC on both sides yields Equation 33. □

Proof of Theorem 20. $m \le f \le M$ and $w \ge 0$ give $m \int w \le \int fw \le M \int w$; so $A := \int fw / \int w \in [m, M]$; by IVT $f(c) = A$. □

3.4. The logarithm and the exponential

Proof of Theorem 21. $\ln(ab) - \ln(b) = \int_b^{ab} \mathrm{d}t/t$; the map $(t, y) \mapsto (t/b, by)$ preserves area and preserves $y = 1/t$, so this equals $\int_1^a \mathrm{d}u/u = \ln a$. □

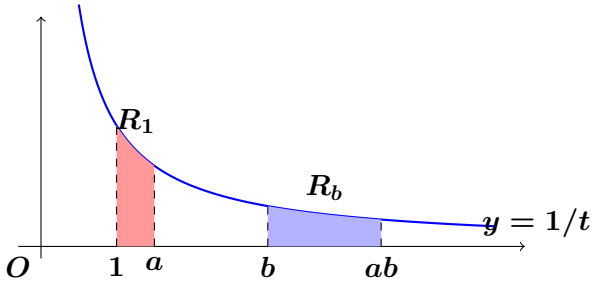


Figure 14. Hyperbolic similarity.

Proof of Theorem 22. $\ln(\exp u \exp v) = \ln(\exp u) + \ln(\exp v) = u + v = \ln(\exp(u+v))$; injectivity of ln gives Equation 36. □

Proof of Theorem 23. Sign-Law gives $\exp(\mathbb{R}_{<0}) =]0,\, 1[, \exp(\{0\}) = \{1\}, \exp(\mathbb{R}_{>0}) =]1,\, \infty[$; Image-Union gives $\exp(\mathbb{R}) = \mathbb{R}_{>0}$. Bijectivity gives $\ln(\mathbb{R}_{>0}) = \mathbb{R}$. □

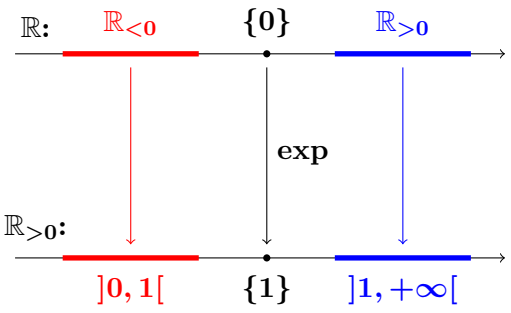


Figure 15. Image of Exponential Function exp

4. Remarks on the geometric development

Barrow's sliding-tangent MVT proof is retained for its historical content; the Barrow-tangency principle Equation 14 is eliminable via Rolle and Monotone + MVT. The FFTC proof uses only area additivity Equation 5 and rectangle comparison Equation 28; no Riemann-sum limit is invoked, as emphasized by Barrow [1] and confirmed in Part 2.

Dependency summary

EVT, IVT → Fermat → Rolle → MVT → Const, Monotone, Cauchy, Darboux, Taylor, Convexity, Inverse. Second FTC from FFTC + Const. IBP, Subst, Weighted MVT from Second FTC or IVT/IntMono. Log from IntAdd + Similarity; Exp from Log; Image from LnBij + Sign-Law + ImgUnion. Barrow-area + MVTInt → FFTC → Log.

Part 2

Reverse-Mathematical Analysis in RCA_0

Every Part 1 entry reappears as a tagged statement with a dependency table and an exact strength in $\{\mathsf{FOL}, \mathsf{RCA}_0, \mathsf{WKL}_0, \mathsf{ACA}_0, \mathsf{ZF}, \mathsf{ZFC}\}$. Geometric proofs are at most WKL_0 as written; §11 gives the matching RCA_0-strength tables via Hardin–Velleman, Yokoyama, or direct argument.

5. Coding conventions

Following Yokoyama [6], we distinguish:

- ***Pointwise differentiability:*** the limit exists for each $x \in I$; no derivative function needed. Hardin–Velleman's Rolle and MVT in RCA_0 use this.
- ***Strong C^k-coding:*** $C^k_{\text{strong}}(I)(f) :\iff \exists g_0, \ldots, g_k : \; g_0 = f \wedge \forall j < k, \; g_{j+1} = g_j'$. All C^k hypotheses in Part 2 mean strong C^k-coding.

- ***Inverse function:*** **Yokoyama [6, Thm. 3.22] in RCA_0 for strong C^1-functions with $f' \neq 0$ on $I, x_0 \in I^\circ$. The pointwise form is not RCA_0-provable.**

6. Research context

6.1. The core hierarchy

RCA_0 = recursive comprehension + Σ^0_1-induction; no choice.

Result	Strength	Reference	Choice
MVT	RCA_0	[3]	none
IVT, EVT	WKL_0	[4]	none
sequential IVT/MVT	$\equiv \mathsf{WKL}_0$	[4, 5]	none
ln/exp chain	RCA_0	[6]	none

Key facts:

- **MVT $\in \mathsf{RCA}_0$ (Hardin–Velleman [3]); no EVT, no IVT.**
- **IVT $\equiv \mathsf{WKL}_0$: sole non-RCA_0 ingredient is $\sup\{x : f(x) < 0\}$.**
- **Inverse function $\in \mathsf{RCA}_0$ (Yokoyama [6, Thm. 3.22]).**
- **ln/exp $\in \mathsf{RCA}_0$: definitional plus integral postulates.**

Remark 1 (Proofs vs. classification).

Part 1** proofs use* **sup, inf, *LUB (ax:R-complete). MVT is RCA_0 via Hardin–Velleman. IVT, EVT are WKL_0-equivalent over RCA_0 and remain WKL_0.

6.2. Illustrative theorems

Theorem	Strength	Reference	Choice
IVT	WKL_0	[4]	none
MVT	RCA_0	[3]	none
Hahn–Banach	$\equiv \mathsf{WKL}_0$	[4]	none
Weierstrass approx.	$\equiv \mathsf{WKL}_0$	[4]	none
Peano existence	$\equiv \mathsf{WKL}_0$	[4]	none
sequential IVT/MVT	$\equiv \mathsf{WKL}_0$	[5]	none
separation	$\equiv \mathsf{ACA}_0$	[15]	none
Banach–Alaoglu	ZFC	[12]	full AC
Tychonoff	ZFC	[13]	full AC
well-ordering	ZFC	[14]	full AC

Strictness: $\mathsf{RCA}_0 \subsetneq \mathsf{WKL}_0 \subsetneq \mathsf{ACA}_0 \subsetneq$ ZFC.

6.3. Dedekind vs. Cauchy completeness

- **Dedekind (LUB): every nonempty bounded-above subset has a supremum. A ZF theorem; equivalent over ordered fields to IVT, EVT, MVT, Darboux [7, 8]. Form used here: ax:R-complete.**

- **Cauchy completeness: every Cauchy sequence converges. ZF-equivalent to $\mathsf{DC}_{\mathbb{R}}$. Not a ZF theorem. Not used here.**

Solovay's model [16]: under ZF + DC, all reals are Lebesgue measurable and AC fails; DC $\Rightarrow$ $\mathsf{DC}_{\mathbb{R}}$, so Cauchy completeness holds there. The choice-free classification is strictly finer than $\mathsf{DC}_{\mathbb{R}}$.

6.4. Constructive RM

In BISH, IVT $\equiv$ LLPO [9]. This document is classical.

6.5. Lean 4 and Mathlib

Lean 4 and Mathlib [10]: about 70.4% of declarations depend on Classical.choice, and 13.1% have choice-free statements with choice-dependent proofs [11]. Our classification is stronger: it certifies choice-free proofs, not just statements.

7. Axioms

The fourteen axioms of $\mathbb{R}$ as a complete ordered field.

Ax 1	ax:add-assoc : $\forall(x,\, y,\, z : \mathbb{R}),\ \ (x + y) + z = x + (y + z)$	Add.Assoc.	ZF
Ax 2	ax:add-comm : $\forall(x,\, y : \mathbb{R}),\ \ x + y = y + x$	Add.Comm.	ZF
Ax 3	ax:add-zero : $\forall(x : \mathbb{R}),\ \ 0 + x = x$	Add.Id.	ZF
Ax 4	ax:add-inv : $\forall(x : \mathbb{R}),\ \ x + (-x) = 0$	Add.Inv.	ZF
Ax 5	ax:mul-assoc : $\forall(x,\, y,\, z : \mathbb{R}),\ \ (x \cdot y) \cdot z = x \cdot (y \cdot z)$	Mul.Assoc.	ZF
Ax 6	ax:mul-comm : $\forall(x,\, y : \mathbb{R}),\ \ x \cdot y = y \cdot x$	Mul.Comm.	ZF
Ax 7	ax:mul-one : $\forall(x : \mathbb{R}),\ \ 1 \cdot x = x$	Mul.Id.	ZF
Ax 8	ax:mul-inv : $\forall(x : \mathbb{R}),$ $(x \neq 0 \to x \cdot x^{-1} = 1) \wedge (x = 0 \to x^{-1} = 0)$	Mul.Inv.	ZF
Ax 9	ax:distrib : $\forall(x,\, y,\, z : \mathbb{R}),\ \ x \cdot (y + z) = x \cdot y + x \cdot z$	Distrib.	ZF
Ax 10	ax:nontriv : $0 \neq 1$	Nontriv.	ZF
Ax 11	ax:order : $\big(\forall x :\ x \leq x\big) \wedge \big(\forall x,$ $y :\ x \leq y \wedge y \leq x \to x = y\big) \wedge \big(\forall x,\, y,$ $z :\ x \leq y \wedge y \leq z \to x \leq z\big) \wedge \big(\forall x,\, y :\ x \leq y \vee y \leq x\big)$	LinearOrder	ZF
Ax 12	ax:add-mon : $\forall(x,\, y,\, z : \mathbb{R}),\ \ x \leq y \to x + z \leq y + z$	AddMon.	ZF
Ax 13	ax:mul-mon : $\forall(x,\, y : \mathbb{R}),\ \ 0 \leq x \wedge 0 \leq y \to 0 \leq x \cdot y$	MulMon.	ZF
Ax 14	ax:complete : $\big(\exists x,\, S(x)\big) \wedge \big(\exists u,\, \forall x,$ $S(x) \to x \leq u\big) \to \exists s,\, \big(\forall x,\, S(x) \to x \leq s\big) \wedge \big(\forall u,\, (\forall x,$ $S(x) \to x \leq u) \to s \leq u\big)$	Cut	ZF

The Cut axiom enters no Part 2 proof; it is ZF-realizable via Dedekind cuts without choice.

8. Postulates

Unproved internal statements. Each is RCA_0-provable or first-order logical; none introduces choice.

8.1. Integral postulates

Pos 1	**Pos:IntLinear** : $\forall(a,b:\mathbb{R}),\ \forall(f,g\in L^1([a,b])),\ \forall(\alpha,\beta:\mathbb{R}),\ \int_a^b(\alpha f(x)+\beta g(x))\,dx=\alpha\int_a^b f(x)\,dx+\beta\int_a^b g(x)\,dx$	**IntLin**	$\mathbf{RCA}_0$
Pos 2	**Pos:IntAdd** : $\forall(a,b,c:\mathbb{R}),\ \forall(f\in L^1([\min(a,b,c),\max(a,b,c)])),\ \int_a^b f(x)\,dx=\int_a^c f(x)\,dx+\int_c^b f(x)\,dx$	**IntAdd**	$\mathbf{RCA}_0$

Note: $f\in L^1([\alpha,\beta])$ and $[\gamma,\delta]\subseteq[\alpha,\beta]$ imply $f\in L^1([\gamma,\delta])$; cited as `restrict` in proof tables.

Pos 3	**Pos:IntMono** : $\forall(a,b:\mathbb{R}),\ a<b\to\forall(f,g\in L^1([a,b])),\ \big(\forall(x\in[a,b]),\ f(x)\le g(x)\big)\to\int_a^b f(x)\,dx\le\int_a^b g(x)\,dx$	**IntMono**	$\mathbf{RCA}_0$
Pos 4	**Pos:IntSubst** : $\forall(a,b:\mathbb{R}),\ \forall(g:\mathbb{R}\to\mathbb{R}),\ \forall(f:\mathbb{R}\to\mathbb{R}),\ g\in C^1([\min(a,b),\max(a,b)])\wedge\big(\forall t\in[\min(a,b),\max(a,b)],\ g'(t)\neq0\big)\wedge f\in C^0(g([\min(a,b),\max(a,b)]))\to\int_{g(a)}^{g(b)}f(x)\,dx=\int_a^b f(g(t))\,g'(t)\,dt$	**IntSubst**	$\mathbf{RCA}_0$
Pos 5	**Pos:ContInt** : $\forall(a,b:\mathbb{R}),\ a<b\to\forall(f:\mathbb{R}\to\mathbb{R}),\ f\in C^0([a,b])\to f\in L^1([a,b])$	**ContInt**	$\mathbf{RCA}_0$
Pos 6	**Pos:Similarity** : $\forall(a,b:\mathbb{R}),\ 0<a\wedge0<b\to\forall(h:\mathbb{R}\to\mathbb{R}),\ \int_b^{a\cdot b}h(t)\,dt=\int_1^a h(bu)\cdot b\,du$	**Sim.**	$\mathbf{RCA}_0$

8.2. Differential postulates

Pos 7	**Pos:ProdRule** : $\forall(f,g:\mathbb{R}\to\mathbb{R}),\ \forall(x:\mathbb{R}),\ \mathsf{Diff}(f,x)\wedge\mathsf{Diff}(g,x)\to\mathsf{Diff}(f\cdot g,x)\wedge(f\cdot g)'(x)=f'(x)g(x)+f(x)g'(x)$	**Prod**	$\mathbf{RCA}_0$
Pos 8	**Pos:ChainRule** : $\forall(f,g:\mathbb{R}\to\mathbb{R}),\ \forall(x:\mathbb{R}),\ \mathsf{Diff}(g,x)\wedge\mathsf{Diff}(f,g(x))\to\mathsf{Diff}(f\circ g,x)\wedge(f\circ g)'(x)=f'(g(x))\cdot g'(x)$	**Chain**	$\mathbf{RCA}_0$
Pos 9	**Pos:InvCont** : $\big(t\mapsto1/t\big)\in C^0(\mathbb{R}_{>0})$	**InvCont**	$\mathbf{RCA}_0$
Pos 10	**Pos:BarrowTangency** : $\forall(G:\mathbb{R}\to\mathbb{R}),\ \forall(x_0,m:\mathbb{R}),\ \mathsf{Diff}(G,x_0)\wedge\Big(\big(\forall(u\neq x_0)\text{ near }x_0,\ G(x_0)+m(u-x_0)\le G(u)\big)\vee\big(\forall(u\neq x_0)\text{ near }x_0,\ G(x_0)+m(u-x_0)\ge G(u)\big)\Big)\to m=G'(x_0)$	**Barrow**	$\mathbf{RCA}_0$

8.3. Logarithmic postulates

Pos 11	**Pos:LnBij** : $\mathsf{Bijective}(\ln:\mathbb{R}_{>0}\to\mathbb{R})$	**LnBij**	$\mathbf{RCA}_0$
Pos 12	**Pos:LnSign** : $\forall(a:\mathbb{R}_{>0}),\ \big(\ln(a)<0\leftrightarrow a<1\big)\wedge\big(\ln(a)=0\leftrightarrow a=1\big)\wedge\big(\ln(a)>0\leftrightarrow a>1\big)$	**LnSign**	$\mathbf{RCA}_0$

8.4. Logical postulate

Pos 13	**Pos:ImgUnion** : $\forall(X,Y:\mathsf{Type}),\ \forall(f:X\to Y),\ \forall(A,B:X\to\mathsf{Prop}),\ f(A\cup B)=f(A)\cup f(B)$	**ImgUnion**	**FOL**

8.5. External results

Pos 14	**Pos:ContImgBdd** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\to\big(\exists L,U:\mathbb{R},\ \forall x:[a,b],\ L\le f(x)\le U\big)$	**BddImg**	$\mathbf{RCA_0}$
Pos 15	**Pos:ContImgAttain** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\to\big(\exists x_{\max}:[a,b],\forall x:[a,b],f(x)\le f(x_{\max})\big)\wedge\big(\exists x_{\min}:[a,b],\forall x:[a,b],f(x_{\min})\le f(x)\big)$	**Attain**	$\mathbf{WKL_0}$

Note on Pos:ContImgBdd and the coding of $C^0([a,b])$. Under the coding of §5 (following Simpson [4]), a continuous function $f:[a,b]\to\mathbb{R}$ carries a coded modulus of continuity, from which boundedness of the image $f([a,b])$ follows in $\mathbf{RCA_0}$. The **Pos:ContImgBdd** tag $\mathbf{RCA_0}$ reflects this standard coding convention. Under an alternative coding — pointwise continuity without a coded modulus — boundedness of continuous functions on $[a,b]$ rises to $\mathbf{WKL_0}$; the choice between codings is a foundation choice, not a mathematical error, and §5 fixes the Simpson convention for this document.

Pos 16	**Pos:Π^0_1Sup** : $\forall(\varphi:\mathbb{N}\times\mathbb{R}\to\{0,1\}),\ \forall(a,b:\mathbb{R}),\ a\le b\to\big(\exists x\in[a,b],\forall n\,\varphi(n,x)=0\big)\to\exists s\in[a,b],\ \big(\forall x\in[a,b],(\forall n\,\varphi(n,x)=0)\to x\le s\big)\wedge\big(\forall u\in[a,b],(\forall x\in[a,b],(\forall n\,\varphi(n,x)=0)\to x\le u\big)\to s\le u\big)$	**Sup**	$\mathbf{WKL_0}$
Pos 17	**Pos:Σ^0_1Choice** : $\big(\forall n,\exists m,R(n,m)\big)\to\exists g,\forall n,R(n,g(n))$	Σ^0_1	$\mathbf{RCA_0}$
Pos 18	**Pos:SeqCont** : $\forall(f:[a,b]\to\mathbb{R}),f\in C^0([a,b])\to\forall x:[a,b],\ \forall(x_n:\mathbb{N}\to[a,b]),\ (x_n\to x)\to\big(f(x_n)\to f(x)\big)$	**SeqCont**	$\mathbf{RCA_0}$
Pos 19	**Pos:HVRolle** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\wedge\big(\forall x:(a,b),\ \mathsf{Diff}(f,x)\big)\wedge f(a)=f(b)\to\exists c:(a,b),\ f'(c)=0$	**HVRolle**	$\mathbf{RCA_0}$
Pos 20	**Pos:YokoyamaInv** : $\forall(f:I\to\mathbb{R}),\ \forall(x_0:I^\circ),\ C^1_{\text{strong}}(I)(f)\wedge(\forall x:I,f'(x)\neq0)\wedge\mathsf{StrictMono}(f)\to\mathsf{Diff}(f^{-1},f(x_0))\wedge(f^{-1})'(f(x_0))=1/f'(x_0)$	**YokInv**	$\mathbf{RCA_0}$

9. Definitions

Definitional abbreviations available in $\mathbf{RCA_0}$: set notation by Δ^0_1-comprehension; L^1, C^k, C^k_{strong} by Σ^0_1 predicates; A^x_a, ln, exp by unique existence + graphs; Diff, ChordSlope, TangentSlope, LocalExt, Increasing, StrictInc, StrictDec, StrictMono by definitional extension.

Def 1	**Def:$\mathsf{R}_{>0}$** : $\mathbb{R}_{>0}:=\{x:\mathbb{R}\mid x>0\}$, $\mathbb{R}_{<0}:=\{x:\mathbb{R}\mid x<0\}$, $\mathbb{R}_{\ge0}:=\{x:\mathbb{R}\mid x\ge0\}$	$\mathsf{R}_{>0}$	$\mathbf{RCA_0}$
Def 2	**Def:L^1** : $L^1([a,b]):=\{f:\mathbb{R}\to\mathbb{R}\mid\int_a^b|f(x)|\,dx<\infty\}$	L^1	$\mathbf{RCA_0}$

Def 3	**Def:C^k** : $\forall(k:\mathbb{N}),\ \forall(U:\mathbb{R}\to\mathsf{Prop}),\ C^k(U):=\{f:\mathbb{R}\to\mathbb{R}\mid f\ \text{k−timescont. diff. on}\ U\}$	C^k	$\mathbf{RCA_0}$
Def 4	**Def:$\mathsf{C}^k_{\text{strong}}$** : $\forall(k:\mathbb{N}),\ \forall(I:\mathbb{R}\to\mathsf{Prop}),\ C^k_{\text{strong}}(I)(f):\iff\exists(g_0,\dots,g_k:\mathbb{R}\to\mathbb{R}),\ g_0=f\wedge\big(\forall j\le k,\ g_j\in C^0(I)\big)\wedge\big(\forall j<k,\ g_{j+1}=(g_j)'\big)$	Strong C^k	$\mathbf{RCA_0}$

Def 5	**Def:Diff** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(x:\mathbb{R}),\ \mathsf{Diff}(f,x) :\iff \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ exists; $\quad f'(x) :=$ that limit	Diff	RCA_0
Def 6	**Def:Area** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(a,x:\mathbb{R}),\ A_a^x := \int_a^x f(t)\,dt$ with $A_x^x=0,\ A_x^{x+\mathrm{d}x}=-A_{x+\mathrm{d}x}^x\ (\mathrm{d}x<0)$	Area	RCA_0
Def 7	**Def:chord** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(a,b:\mathbb{R}),\ a\neq b\to \mathsf{ChordSlope}(f,a,b) := \frac{f(b)-f(a)}{b-a}$	Chord	RCA_0
Def 8	**Def:tangent** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(c:\mathbb{R}),\ \mathsf{Diff}(f,c)\to \mathsf{TangentSlope}(f,c) := f'(c)$	Tangent	RCA_0
Def 9	**Def:LocExt** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(c:\mathbb{R}),\ \mathsf{LocalExt}(f,c) :\iff \exists(\varepsilon>0),\ \big(\forall(x\in(c-\varepsilon,c+\varepsilon)),\ f(x)\le f(c)\big)\vee\big(\forall(x\in(c-\varepsilon,c+\varepsilon)),\ f(x)\ge f(c)\big)$	LocExt	RCA_0
Def 10	**Def:Increasing** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{Increasing}(f) :\iff \forall(x,y:\mathbb{R}),\ x<y\to f(x)\le f(y)$	Incr	RCA_0
Def 11	**Def:StrictInc** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{StrictInc}(f) :\iff \forall(x,y:\mathbb{R}),\ x<y\to f(x)<f(y)$	SIncr	RCA_0
Def 12	**Def:StrictDec** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{StrictDec}(f) :\iff \forall(x,y:\mathbb{R}),\ x<y\to f(x)>f(y)$	SDecr	RCA_0
Def 13	**Def:StrictMono** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{StrictMono}(f) :\iff \mathsf{StrictInc}(f)\vee\mathsf{StrictDec}(f)$	SMono	RCA_0
Def 14	**Def:ln** : $\forall(x:\mathbb{R}_{>0}),\ \ln(x) := \int_1^x \frac{1}{t}\,dt$	ln	RCA_0
Def 15	**Def:exp** : $\exp := \ln^{-1} : \mathbb{R}\to\mathbb{R}_{>0}$	exp	RCA_0

10. Statements

Statements and their strengths. Proofs in §11. All C^k hypotheses are strong C^k-coding (§5); pointwise differentiability appears only in Hardin–Velleman.

10.1. Topological theorems

Thm 1	**EVT** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\to\exists x_{\max}\in[a,b],\ \exists x_{\min}\in[a,b],\ \forall(x\in[a,b]),\ f(x_{\min})\le f(x)\le f(x_{\max})$	EVT	WKL_0
Thm 2	**IVT** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\wedge f(a)\cdot f(b)<0\to\exists(c\in(a,b)),\ f(c)=0$	IVT	WKL_0

10.2. Differentiation theorems

Thm 3	**Fermat** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(a:\mathbb{R}),\ \forall(b:\mathbb{R}),\ \forall(c:\mathbb{R}),\ a<c\wedge c<b\wedge\mathsf{LocalExt}(f,c)\wedge\mathsf{Diff}(f,c)\to f'(c)=0$	Fermat	RCA_0
Thm 4	**Rolle** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\wedge f\in C^1((a,b))\wedge f(a)=f(b)\to\exists(c\in(a,b)),\ f'(c)=0$	Rolle	RCA_0
Thm 5	**MVT** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\wedge f\in C^1((a,b))\to\exists(c\in(a,b)),\ f'(c)=\mathsf{ChordSlope}(f,a,b)$	MVT	RCA_0
Thm 6	**Const** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\wedge f\in C^1((a,b))\wedge(\forall x\in(a,b),\ f'(x)=0)\to(\forall x\in[a,b],\ f(x)=f(a))$	Const	RCA_0

Thm 7	**Monotone** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\wedge f\in C^1((a,b))\to\big((\forall x\in(a,b),\ f'(x)\geq 0)\to \mathsf{Increasing}(f)\big)\wedge\big((\forall x\in(a,b),\ f'(x)>0)\to\mathsf{StrictInc}(f)\big)$	**Mono**	$\mathbf{RCA_0}$
Thm 8	**CauchyMVT** : $\forall(f,g:[a,b]\to\mathbb{R}),\ f,g\in C^0([a,b])\wedge f,g\in C^1((a,b))\wedge(\forall x\in(a,b),\ g'(x)\neq 0)\to\exists(c\in(a,b)),\ \frac{f'(c)}{g'(c)}=\frac{f(b)-f(a)}{g(b)-g(a)}$	**Cauchy**	$\mathbf{RCA_0}$
Thm 9	**Darboux** : $\forall(f:[a,b]\to\mathbb{R}),\ \forall(\lambda:\mathbb{R}),\ \mathsf{Diff}(f,a)\wedge\mathsf{Diff}(f,b)\wedge\big(\forall x\in(a,b),\ \mathsf{Diff}(f,x)\big)\wedge f'(a)<\lambda<f'(b)\to\exists(c\in(a,b)),\ f'(c)=\lambda$	**Darboux**	$\mathbf{WKL_0}$
Thm 10	**LHopital** : $\forall(f,g:\mathbb{R}\to\mathbb{R}),\ \forall(a:\mathbb{R}),\ \forall(L:\mathbb{R}),\ f(a)=g(a)=0\wedge\big(\forall x\text{ near }a,\ \mathsf{Diff}(f,x)\wedge\mathsf{Diff}(g,x)\wedge g'(x)\neq 0\big)\wedge\lim_{x\to a}\frac{f'(x)}{g'(x)}=L\to\lim_{x\to a}\frac{f(x)}{g(x)}=L$	**L'Hôpital**	$\mathbf{RCA_0}$
Thm 11	**Taylor** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(a,x:\mathbb{R}),\ \forall(n:\mathbb{N}),\ a\neq x\wedge C^{n+1}_{\mathbf{strong}}([\min(a,x),\max(a,x)])(f)\to\exists(c\in(\min(a,x),\max(a,x))),\ f(x)=\sum_{k=0}^{n}\frac{f^{(k)}(a)}{k!}(x-a)^k+\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$	**Taylor**	$\mathbf{RCA_0}$
Thm 12	**Convexity** : $\forall(f:[a,b]\to\mathbb{R}),\ C^2_{\mathbf{strong}}([a,b])(f)\wedge(\forall x\in(a,b),\ f''(x)\geq 0)\to\forall(x,x_0\in[a,b]),\ f(x)\geq f(x_0)+f'(x_0)(x-x_0)$	**Convex**	$\mathbf{RCA_0}$
Thm 13	**Inverse** : $\forall(f:I\to\mathbb{R}),\ \forall(x_0\in I^\circ),\ C^1_{\mathbf{strong}}(I)(f)\wedge(\forall x\in I,\ f'(x)\neq 0)\wedge\mathsf{StrictMono}(f)\to\mathsf{Diff}(f^{-1},f(x_0))\wedge(f^{-1})'(f(x_0))=1/f'(x_0)$	**Inv**	$\mathbf{RCA_0}$

10.3. Integration theorems

Lem 1	**BarrowArea** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\wedge\mathsf{Increasing}(f)\to\forall(x_0\in[a,b]),\ \forall(\mathrm{d}x\neq 0,\ x_0+\mathrm{d}x\in[a,b]),\ f(x_0)\,\mathrm{d}x\leq A_{x_0}^{x_0+\mathrm{d}x}\leq f(x_0+\mathrm{d}x)\,\mathrm{d}x$	**BarrowArea**	$\mathbf{RCA_0}$
Lem 2	**MVTInt** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\to\forall(x_0,\mathrm{d}x\neq 0),\ x_0+\mathrm{d}x\in[a,b]\to\exists(\theta\text{ between }x_0,x_0+\mathrm{d}x),\ A_{x_0}^{x_0+\mathrm{d}x}=f(\theta)\,\mathrm{d}x$	**MVTInt**	$\mathbf{RCA_0}$
Thm 14	**FFTC** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b])\to\forall(x\in(a,b)),\ \frac{d}{dx}A_a^x=f(x)$	**FFTC**	$\mathbf{RCA_0}$
Thm 15	**SecondFTC** : $\forall(F:[a,b]\to\mathbb{R}),\ F\in C^1([a,b])\to\int_a^b F'(t)\,dt=F(b)-F(a)$	**SecondFTC**	$\mathbf{RCA_0}$
Thm 16	**IBP** : $\forall(f,g\in C^1([a,b])),\ \int_a^b f\,g'=[fg]_a^b-\int_a^b f'\,g$	**IBP**	$\mathbf{RCA_0}$
Thm 17	**Subst** : $\forall(g\in C^1([a,b])),\ \forall(f\in C^0(g([a,b]))),\ \int_{g(a)}^{g(b)}f(u)\,du=\int_a^b f(g(t))\,g'(t)\,dt$	**Subst**	$\mathbf{RCA_0}$
Thm 18	**WeightedMVT** : $\forall(f\in C^0([a,b])),\ \forall(w\geq 0\text{ Riemann integrable}),\ \int_a^b w>0\to\exists(c\in[a,b]),\ \int_a^b f\,w=f(c)\int_a^b w$	**WMVT**	$\mathbf{WKL_0}$

10.4. The logarithm and the exponential

Lem 3	**PosClosedMul** : $\forall(a,b:\mathbb{R}),\ 0<a\wedge 0<b\to 0<a\cdot b$	**Pos Closed**	$\mathbf{RCA_0}$
Lem 4	**LogProperty** : $\forall(a,b:\mathbb{R}_{>0}),\ \ln(a)+\ln(b)=\ln(a\cdot b)$	**LogAdd**	$\mathbf{RCA_0}$

Lem 5	ExpProperty : $\forall(u, v : \mathbb{R}),\ \exp(u) \cdot \exp(v) = \exp(u + v)$	ExpAdd	RCA_0
Thm 19	ImageExpLn : $\exp(\mathbb{R}) = \mathbb{R}_{>0} \wedge \ln(\mathbb{R}_{>0}) = \mathbb{R}$	Image	RCA_0

10.5. Illustrative theorems

Theorem	Strength	Choice
IVT	WKL_0	none
MVT	RCA_0	none
Hahn–Banach	WKL_0	none
Weierstrass	WKL_0	none
Peano	WKL_0	none
separation	ACA_0	none
Banach–Alaoglu	ZFC	full AC
Tychonoff	ZFC	full AC

Remark 2 (EVT tag).
EVT $\equiv$ WKL_0 over RCA_0 [4]. Entries whose printed proofs invoke EVT transitively: Rolle, MVT, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity, Darboux, WeightedMVT; Part 1 proofs are at most WKL_0 as written. Rolle and MVT are RCA_0 via Hardin–Velleman [3]; the descendants inherit RCA_0 under strong-C^k-coding. Darboux and WeightedMVT are WKL_0 as stated. Fermat, FFTC, SecondFTC, IBP, Subst, BarrowArea, MVTInt, Inverse, and the ln/exp *chain are RCA_0 (direct; [3, 6] for Inverse and MVTInt). IVT is WKL_0-equivalent for the* sup *form; the neighbourhood argument below uses only continuity.*

11. Proofs

Starred subsections, one per proof, in statement order. Each proof is a four-column dependency table: first row **Hyp**, last row $\to$ **I**$(1, \ldots, N-1)$. Justifications use only **Hyp**, **def**, **intro**, **Ax:**, **Pos:**, **Def:**, **Lem:**, **Thm:**, **Eq**$(\ldots)$, or a logical rule ($\to$ **I**, $\wedge$**E**, $\vee$**E**, $\forall$**I**, $\forall$**E**, $\exists$**I**, $\exists$**E**, $\neg$**I**, $\neg$**E**, $\bot$**I**, $\bot$**E**, **RAA**, **DNE**, **LEM**). No external citation appears inside a proof table.

Proof of Theorem 1 (EVT)

1	$f \in C^0([a, b])$	hypothesis	Hyp
2	$\exists L, U,\ \forall x,\ L \le f(x) \le U$	bdd	Pos:14(1)
3	$\exists x_{\max},\ \forall x,\ f(x) \le f(x_{\max})$	max	Pos:15(1)
4	$\exists x_{\min},\ \forall x,\ f(x_{\min}) \le f(x)$	min	Pos:15(1)
5	$\forall x,\ f(x_{\min}) \le f(x) \le f(x_{\max})$	combine	Eq(3, 4)
6	$\top$	close	$\to$ I$(1, \ldots, 5)$

Proof of Theorem 2 (IVT)

1	$f \in C^0([a,\,b]) \wedge f(a)f(b) < 0$	hypothesis	Hyp
2	$f(a) < 0 < f(b)$	sign	Eq(1)
3	$S := \{x \in [a,b] : f(x) < 0\}$ is Σ^0_1	def	Eq(2)
4	$P := \{x \in [a,b] : -f(x) \le 0\}$ is Π^0_1	def	Eq(2)
5	$c := \sup S = -\inf P$	sup	Pos:16(4)
6	$\forall n\, \exists x \in S,\ c - \frac{1}{n+1} < x \le c$	sup	Eq(5)
7	$\exists (x_n) \subset S,\ x_n \to c$ (codes of reals)	Σ^0_1-choice	Pos:17(6)
8	$f(x_n) \to f(c)$	seq	Pos:18(1, 7)
9	$f(c) \le 0$	lim	Eq(7, 8)
10	$f(c) < 0 \to \exists \delta,\ \forall u \in (c-\delta,\, c+\delta),\ \ f(u) < 0$	cont	Def:3(1)
11	$f(c) < 0 \to (c + \delta/2 \in S \wedge c < c + \delta/2)$	shift	Eq(3, 10)
12	$f(c) < 0 \to \bot$	contra	Eq(5, 11)
13	$f(c) \ge 0$	RAA	Eq(12)
14	$f(c) = 0$	antisym	Eq(9, 13)
15	$\top$	close	$\to$ I(1, . . . , 14)

Coding note (IVT). In row 3, $S = \{x \in [a,b] : f(x) < 0\}$ is read as a Σ^0_1 set of *codes* for reals, so Pos:Σ^0_1Choice (row 7) applies at the level of codes. In row 4, $P = \{x \in [a,b] : -f(x) \le 0\}$ is the Π^0_1 complement of S in $[a,b]$; Pos:Π^0_1Sup applied to P yields $\inf P$, and $\sup S = -\inf P$ (cf. Simpson [4, §IV.2]).

Proof of Theorem 3 (Fermat)

1	local max, Diff(f, c)	hypothesis	Hyp
2	$\forall h > 0$ small, $(f(c+h) - f(c))/h \le 0$	locmax	Def:9(1)
3	$f'(c) \le 0$	osl	Def:5(2)
4	$\forall h < 0$ small, $(f(c+h) - f(c))/h \ge 0$	locmax	Def:9(1)
5	$f'(c) \ge 0$	osl	Def:5(4)
6	$f'(c) = 0$	antisym	Ax:11(3, 5)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Theorem 4 (Rolle)

1	$f \in C^0 \wedge f \in C^1 \wedge f(a) = f(b)$	hypothesis	Hyp
2	$\exists c,\ \ f'(c) = 0$	HVRolle	Pos:19(1)
3	$\top$	close	$\to$ I(1, 2)

Proof of Theorem 5 (MVT)

1	$f \in C^0 \wedge f \in C^1$	hypothesis	Hyp
2	$L :=$ chord, $g := f - L$	def	Def:7(1)
3	$g \in C^0 \wedge g \in C^1 \wedge g(a) = g(b) = 0$	construct	Eq(2)
4	$\exists c,\ g'(c) = 0$	HVRolle	Pos:19(3)
5	$g'(c) = f'(c) -$ ChordSlope	diff	Eq(2)
6	$f'(c) =$ ChordSlope	combine	Eq(4, 5)
7	$\top$	close	$\to$ I(1, ..., 6)

Proof of Theorem 6 (Const)

1	$f \in C^0 \wedge f \in C^1 \wedge f' \equiv 0$	hypothesis	Hyp
2	$\forall x,\ \exists c,\ f(x) - f(a) = f'(c)(x - a)$	MVT	Thm:5(1)
3	$f'(c)(x - a) = 0$	combine	Eq(1, 2)
4	$f(x) = f(a)$	combine	Eq(2, 3)
5	$\top$	close	$\to$ I(1, ..., 4)

Proof of Theorem 7 (Monotone)

1	$f \in C^0 \wedge f \in C^1$	hypothesis	Hyp
2	$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$	MVT	Thm:5(1)
3	$f' \geq 0 \to f(x_2) \geq f(x_1)$	sign	Eq(1, 2)
4	$f' > 0 \to f(x_2) > f(x_1)$	sign	Eq(1, 2)
5	$f' \geq 0 \to$ Increasing	combine	Def:10(3)
6	$f' > 0 \to$ StrictInc	combine	Def:11(4)
7	$\top$	close	$\to$ I(1, ..., 6)

Proof of Theorem 8 (CauchyMVT)

1	$f, g \in C^0 \wedge f, g \in C^1 \wedge g' \neq 0$	hypothesis	Hyp
2	$h := (f(b) - f(a))g - (g(b) - g(a))f$	def	def
3	$h \in C^0 \wedge h \in C^1 \wedge h(a) = h(b)$	construct	Eq(2)
4	$\exists c,\ h'(c) = 0$	HVRolle	Pos:19(3)
5	$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$	diff	Eq(2, 4)
6	$g'(c) \neq 0$	hyp	Eq(1, 4)
7	$g(b) \neq g(a)$	Rolle(g)	Thm:4(1)
8	$f'(c)/g'(c) = (f(b) - f(a))/(g(b) - g(a))$	divide	Eq(5, 6, 7)
9	$\top$	close	$\to$ I(1, ..., 8)

Proof of Theorem 9 (Darboux)

1	f' exists, $f'(a) < \lambda < f'(b)$	hypothesis	Hyp
2	$g := f - \lambda x,\ g'(a) < 0 < g'(b)$	def	Eq(1)

3	$\exists h > 0,\ g(a+h) < g(a)$	$g'(a) < 0$	Def:5(2)
4	$\exists h > 0,\ g(b-h) < g(b)$	$g'(b) > 0$	Def:5(2)
5	$\exists c \in [a, b],\ \forall x,\ g(c) \leq g(x)$	EVT	Thm:1(1)
6	$c \neq a \wedge c \neq b$	combine	Eq(3, 4, 5)
7	$c \in (a, b)$	combine	Eq(5, 6)
8	$g'(c) = 0$	Fermat	Thm:3(7)
9	$f'(c) = \lambda$	def	Eq(2, 8)
10	$\top$	close	$\rightarrow$ I(1, ..., 9)

Proof of Theorem 10 (LHopital)

1	$f(a) = g(a) = 0 \wedge (\forall x \text{ near } a,\ g'(x) \neq 0) \wedge \lim_{x \to a} f'/g' = L$	hypothesis	Hyp
2	$g \neq 0$ near a	Rolle	Thm:4(1)
3	$f(x)/g(x) = f'(c_x)/g'(c_x)$	Cauchy	Thm:8(1, 2)
4	$x \to a^+ \Rightarrow c_x \to a^+$	combine	Eq(3)
5	$f'/g'(c_x) \to L$	combine	Eq(1, 4)
6	$f/g(x) \to L\ (a^+)$	combine	Eq(3, 5)
7	$f/g(x) \to L\ (a^-)$ sym.	Cauchy	Thm:8(1, 2)
8	$\lim_{x \to a} f/g = L$	combine	Eq(6, 7)
9	$\top$	close	$\rightarrow$ I(1, ..., 8)

Proof of Taylor 11 (Taylor)

1	$f \in C^{n+1}$	hypothesis	Hyp
2	$P :=$ Taylor poly	def	def
3	$M := (f(x) - P(x))/(x-a)^{n+1}$	def	Eq(1, 2)
4	$h := f - P - M(t-a)^{n+1}$	def	Eq(2, 3)
5	$h^{(j)}(a) = 0\ (j \leq n)$	def	Eq(2, 4)
6	$h(x) = 0$	def	Eq(3, 4)
7	$h \in C^n \wedge h \in C^{n+1}$	smooth	Eq(1, 4)
8	$\exists c_1,\ h'(c_1) = 0$	HVRolle	Pos:19(7)
9	$h'(a) = h'(c_1) = 0$	combine	Eq(5, 8)
10	$\exists c_2,\ h''(c_2) = 0$	HVRolle	Pos:19(7)
11	(iterate $n+1$ times)	HVRolle	Pos:19(7)
12	$\exists c,\ h^{(n+1)}(c) = 0$	combine	Eq(8, ..., 11)
13	$h^{(n+1)} = f^{(n+1)} - M(n+1)!$	def	Eq(2, 4)
14	$M = f^{(n+1)}(c)/(n+1)!$	combine	Eq(12, 13)
15	$f(x) =$ Taylor sum $+ \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$	combine	Eq(2, 3, 14)
16	$\top$	close	$\rightarrow$ I(1, ..., 15)

Proof of Theorem 12 (Convexity)

1	$f \in C^2 \land f'' \geq 0$	hypothesis	Hyp
2	$g := f - f(x_0) - f'(x_0)(x - x_0)$	def	Def:8(1)
3	$g'' = f'' \geq 0$	diff	Eq(1, 2)
4	g' nondecreasing, $g'(x_0) = 0$	Monotone	Thm:7(3)
5	$g' \leq 0$ $(x < x_0)$, $g' \geq 0$ $(x > x_0)$	split	Eq(4)
6	g nonincr./nondecr.	Monotone	Thm:7(5)
7	$g(x_0) = 0$	def	Eq(2)
8	$g \geq 0$	combine	Eq(6, 7)
9	$f(x) \geq f(x_0) + f'(x_0)(x - x_0)$	combine	Eq(2, 8)
10	$\top$	close	$\to$ I(1, ..., 9)

Proof of Theorem 13 (Inverse)

1	$x_0 \in I^\circ \land C^1_{\text{strong}}(I)(f) \land (\forall x : I,\ f'(x) \neq 0) \land \mathsf{StrictMono}(f)$	hypothesis	Hyp
2	$\mathsf{Diff}(f^{-1}, f(x_0)) \land (f^{-1})'(f(x_0)) = 1/f'(x_0)$	Yokoyama	Pos:20(1)
3	$\top$	close	$\to$ I(1, 2)

Proof of Lemma 1 (BarrowArea)

1	$f \in C^0([a, b])$, $\mathsf{Incr}(f)$, $x_0 < x_0 + \mathrm{d}x$	hypothesis	Hyp
2	$f \in L^1([a, b])$	cont	Pos:5(1)
3	$f(x_0) \leq f \leq f(x_0 + \mathrm{d}x)$	incr	Def:10(1)
4	inscribed $\leq A \leq$ circumscribed	mono	Pos:3(2, 3)
5	$f(x_0)\mathrm{d}x \leq A \leq f(x_0 + \mathrm{d}x)\mathrm{d}x$	rect	Eq(4)
6	$\mathrm{d}x < 0 :\ A = -A'$	orient	Def:6
7	same inequality	negate	Eq(6)
8	$\top$	close	$\to$ I(1, ..., 7)

Proof of Lemma 2 (MVTInt)

1	$f \in C^0$, $x_0, x_0 + \mathrm{d}x \in [a, b]$	hypothesis	Hyp
2	$G(t) := A^t_{x_0}$	area	Def:6(1)
3	$G \in C^0 \land G' = f$	FFTC	Thm:14(1)
4	$\exists \theta,\ G'(\theta) = (G(x_0 + \mathrm{d}x) - G(x_0))/\mathrm{d}x$	MVT	Thm:5(1, 3)
5	$G(x_0) = 0 \land G(x_0 + \mathrm{d}x) = A$	def	Def:6(2)
6	$A = f(\theta)\mathrm{d}x$	combine	Eq(3, 4, 5)
7	$\top$	close	$\to$ I(1, ..., 6)

Proof of Theorem 14 (FFTC)

1	$f \in C^0,\ x_0 \in (a,\, b),\ \varepsilon > 0$	hypothesis	Hyp
2	$f \in L^1([a,\, b])$	cont	Pos:5(1)
3	$f \in L^1([\min(a, x_0, x_0+h), \max(a, x_0, x_0+h)])$	restrict	Eq(2)
4	$A_a^{x_0+h} = A_a^{x_0} + A_{x_0}^{x_0+h}$	additivity	Pos:2(3)
5	$\exists m_\varepsilon,\, M_\varepsilon,\ \forall u \in I_h,\ m_\varepsilon \le f(u) \le M_\varepsilon,\ \lvert m_\varepsilon - f(x_0)\rvert,$ $\lvert M_\varepsilon - f(x_0)\rvert < \varepsilon$	cont	Def:3(1)
6	$h > 0:\ m_\varepsilon h \le A \le M_\varepsilon h$	mono	Pos:3(3, 5)
7	$h < 0:\ M_\varepsilon h \le A \le m_\varepsilon h$	orient	Def:6
8	$m_\varepsilon \le (A_a^{x_0+h} - A_a^{x_0})/h \le M_\varepsilon$	divide	Eq(4, 6, 7)
9	$\mathrm{d}A_a^x/\mathrm{d}x\rvert_{x_0} = f(x_0)$	squeeze	Eq(5, 8)
10	x_0 arbitrary	gen	Eq(9)
11	$\top$	close	$\to$ I(1, ..., 10)

Proof of Theorem 15 (SecondFTC)

1	$F \in C^1,\ f := F' \in C^0$	hypothesis	Hyp
2	$G(x) := \int_a^x f,\ G' = f$	FFTC	Thm:14(1)
3	$(F - G)' = 0$	diff	Eq(1, 2)
4	$F - G$ constant	Const	Thm:6(3)
5	$F(b) - F(a) = \int_a^b f$	combine	Eq(4)
6	$\top$	close	$\to$ I(1, ..., 5)

Proof of Theorem 16 (IBP)

1	$f,\, g \in C^1$	hypothesis	Hyp
2	$f'g,\, fg' \in C^0([a,\, b]) \subseteq L^1([a,\, b])$	cont	Pos:5(1)
3	$(fg)' = f'g + fg'$	prod	Pos:7(1)
4	$\int_a^b (fg)' = [fg]_a^b$	SecondFTC	Thm:15(2)
5	$\int (f'g + fg') = \int f'g + \int fg'$	linear	Pos:1(2, 4)
6	$[fg]_a^b = \int f'g + \int fg'$	combine	Eq(4, 5)
7	$\int fg' = [fg]_a^b - \int f'g$	rearrange	Eq(6)
8	$\top$	close	$\to$ I(1, ..., 7)

Proof of Theorem 17 (Subst)

1	$g \in C^1,\ f \in C^0(g([a,\, b]))$	hypothesis	Hyp
2	$f \in L^1(g([a,\, b])) \wedge f(g)g' \in L^1([a,\, b])$	cont	Pos:5(1)
3	$F := \int_{g(a)}^{\cdot} f,\ F' = f$	FFTC	Thm:14(1)
4	$(F \circ g)' = f(g)g'$	chain	Pos:8(1, 3)

5	$\int f(g)g' = F(g(b)) - F(g(a))$	**SecondFTC**	**Thm:15(4)**
6	$\int_{g(a)}^{g(b)} f = F(g(b)) - F(g(a))$	**SecondFTC**	**Thm:15(3)**
7	both $= F(g(b)) - F(g(a))$	**combine**	**Eq(5, 6)**
8	$\top$	**close**	**$\to$ I(1, ..., 7)**

Proof of Theorem 18 (WeightedMVT)

1	$f \in C^0$, $w \geq 0$ int., $\int w > 0$	**hypothesis**	**Hyp**
2	f, $fw \in L^1([a, b])$	**int**	**Pos:5(1)**
3	$m := \min f$, $M := \max f$	**EVT**	**Thm:1(1)**
4	$m \int w \leq \int fw \leq M \int w$	**mono**	**Pos:3(2, 3)**
5	$A := \int fw / \int w \in [m, M]$	**divide**	**Eq(4)**
6	$\exists c$, $f(c) = A$	**IVT**	**Thm:2(1, 5)**
7	$\int fw = f(c) \int w$	**combine**	**Eq(5, 6)**
8	$\top$	**close**	**$\to$ I(1, ..., 7)**

Proof of Lemma 3 (PosClosedMul)

1	$0 < a$, $0 < b$	**hypothesis**	**Hyp**
2	$0 \leq a$, $0 \leq b$	**weaken**	**Ax:11(1)**
3	$0 \leq ab$	**mulMon**	**Ax:13(2)**
4	$a \neq 0$, $b \neq 0$	**combine**	**Ax:11(2)**
5	$ab \neq 0$	**combine**	**Ax:8(4)**
6	$0 < ab$	**combine**	**Eq(3, 5)**
7	$\top$	**close**	**$\to$ I(1, ..., 6)**

Proof of Lemma 4 (LogProperty)

1	$a, b > 0$	**hypothesis**	**Hyp**
2	$ab > 0$	**closed**	**Lem:3(1)**
3	$(t \mapsto 1/t) \in C^0(\mathbb{R}_{>0})$	**InvCont**	**Pos:9**
4	$(t \mapsto 1/t) \in L^1([\min(1, b, ab), \max(1, b, ab)])$	**ContInt**	**Pos:5(3)**
5	$\int_1^{ab} \frac{\mathrm{d}t}{t} = \int_1^b \frac{\mathrm{d}t}{t} + \int_b^{ab} \frac{\mathrm{d}t}{t}$	**additivity**	**Pos:2(4)**
6	$\ln(ab) = \int_1^{ab} \frac{\mathrm{d}t}{t}$, $\ln(b) = \int_1^b \frac{\mathrm{d}t}{t}$	**def**	**Def:14(1, 2)**
7	$\ln(ab) - \ln(b) = \int_b^{ab} \frac{\mathrm{d}t}{t}$	**combine**	**Eq(5, 6)**
8	$g := (u \mapsto bu) \in C^1([\min(1, a), \max(1, a)])$, $g' = b \neq 0$	**lin**	**Ax:6(1, 2)**
9	$\int_b^{ab} \frac{\mathrm{d}t}{t} = \int_1^a \frac{1}{bu}\, b\, \mathrm{d}u$	**subst**	**Pos:4(3, 8)**
10	$\frac{1}{bu} \cdot b = \frac{1}{u}$	**field**	**Ax:8(1, 2)**

11	$\int_1^a \frac{1}{bu}\, b\, \mathrm{d}u = \int_1^a \frac{\mathrm{d}u}{u}$	combine	Eq(10)
12	$\int_1^a \frac{\mathrm{d}u}{u} = \ln(a)$	def	Def:14(1)
13	$\ln(ab) - \ln(b) = \ln(a)$	combine	Eq(7, 9, 11, 12)
14	$\ln(a) + \ln(b) = \ln(ab)$	rearrange	Eq(13)
15	$\top$	close	$\to$ I(1, . . . , 14)

Proof of Lemma 5 (ExpProperty)

1	$u,\, v \in \mathbb{R}$	hypothesis	Hyp
2	$a := \exp u,\ \ b := \exp v$	def	def
3	$a,\, b > 0$	def	Def:15(2)
4	$\ln a = u,\ \ \ln b = v$	inv	Def:15(2, 3)
5	$ab > 0$	closed	Lem:3(3)
6	$\ln(ab) = \ln a + \ln b = u + v$	Log	Lem:4(3, 4, 5)
7	$\exp(u + v) = \exp(\ln(ab)) = ab$	inv	Def:15(2, 6)
8	$ab = \exp u \cdot \exp v$	def	Eq(2)
9	$\exp u \cdot \exp v = \exp(u + v)$	combine	Eq(7, 8)
10	$\top$	close	$\to$ I(1, . . . , 9)

Citation note (ExpProperty). Row 5 cites **Lem:PosClosedMul(3):** current-table row 3 $(a, b > 0)$ supplies the positivity input to **PosClosedMul**, yielding $ab > 0$ at row 5. Row 6 cites **Lem:LogProperty(3,4,5):** current-table row 3 $(a, b > 0)$ supplies **LogProperty**'s positivity precondition; row 4 $(\ln a = u,\, \ln b = v)$ supplies the logarithmic values; row 5 $(ab > 0)$ makes $\ln(ab)$ well-typed in **LogProperty**'s conclusion. Including row 3 is explicit rather than redundant: **LogProperty**'s hypothesis $a, b : \mathbb{R}_{>0}$ is satisfied by row 3, even though $a =_{\mathrm{def}} \exp u$ already forces $a > 0$.

Proof of Theorem 19 (ImageExpLn)

1	$\forall u,\ \ u < 0 \vee u \geq 0$	tot	Ax:11(1)
2	$\mathbb{R} = \mathbb{R}_{<0} \cup \mathbb{R}_{\geq 0}$	part	Def:1(1)
3	$\exp(\mathbb{R}) = \exp(\mathbb{R}_{<0}) \cup \exp(\mathbb{R}_{\geq 0})$	ImgUnion	Pos:13(2)
4	$\ln(\exp u) = u$	inv	Def:15(1, 4)
5	$\exp(u) < 1 \wedge \exp(u) > 0\ (u < 0)$	Sign	Pos:12(4, 3)
6	$\exp(\mathbb{R}_{<0}) =]0,\, 1[$	bij	Pos:11(5)
7	$\exp(u) \geq 1 \wedge \exp(u) > 0\ (u \geq 0)$	Sign	Pos:12(7, 3)
8	$\exp(\mathbb{R}_{\geq 0}) = [1,\, \infty[$	bij	Pos:11(8)
9	$\exp(\mathbb{R}) =]0,\, \infty[$	combine	Eq(3, 6, 8)
10	$\exp(\mathbb{R}) = \mathbb{R}_{>0}$	def	Def:1(9)
11	$\ln(\mathbb{R}_{>0}) = \mathbb{R}$	bij	Pos:11(1)
12	$\top$	close	$\to$ I(1, . . . , 11)

12. Scripts

Each tree lists entries used by a proof, transitively closed, with strengths. `L0` is the root.

12.1. Foundational leaves

Axioms (all ZF).

```
ax:R-add-assoc      (ZF)
ax:R-add-comm       (ZF)
ax:R-add-zero       (ZF)
ax:R-add-inv        (ZF)
ax:R-mul-assoc      (ZF)
ax:R-mul-comm       (ZF)
ax:R-mul-one        (ZF)
ax:R-mul-inv        (ZF)
ax:R-distrib        (ZF)
ax:R-nontriv        (ZF)
ax:R-order          (ZF)
ax:R-add-monotone   (ZF)
ax:R-mul-monotone-pos (ZF)
ax:R-complete       (ZF, Cut)
```

Postulates.

```
Integral:
Pos:IntLinear       (RCA0)
Pos:IntAdd          (RCA0)
Pos:IntMono         (RCA0)
Pos:IntSubst        (RCA0)
Pos:ContInt         (RCA0)
Pos:Similarity      (RCA0)

Differential:
Pos:ProdRule        (RCA0)
Pos:ChainRule       (RCA0)
Pos:InvCont         (RCA0)
Pos:BarrowTangency (RCA0)

Log/Logic:
Pos:LnBij           (RCA0)
Pos:LnSign          (RCA0)
Pos:ImgUnion        (FOL)

External:
Pos:ContImgBdd      (RCA0)
Pos:ContImgAttain   (WKL0)
Pos:Pi01Sup         (WKL0)
Pos:Sigma01Choice   (RCA0)
Pos:SeqCont         (RCA0)
Pos:HVRolle         (RCA0)
Pos:YokoyamaInv     (RCA0)
```

12.2. Per-result trees

Lemma **PosClosedMul** (**RCA$_0$**)

```
L0 | PosClosedMul                    (RCA0)
L1 |   ax:R-order                    (ZF)
L1 |   ax:R-mul-monotone-pos          (ZF)
L1 |   ax:R-mul-inv                   (ZF)
```

Lemma **LogProperty** (**RCA$_0$**)

```
L0 | LogProperty                     (RCA0)
L1 |   PosClosedMul                  (RCA0)
L1 |   Pos:IntLinear                 (RCA0)
L1 |   Pos:IntAdd                    (RCA0)
L1 |   Pos:IntSubst                  (RCA0)
L1 |   Def:ln                        (RCA0)
L1 |   ax:R-mul-comm                 (ZF)
L1 |   ax:R-mul-inv                  (ZF)
```

Lemma **ExpProperty** (**RCA$_0$**)

```
L0 | ExpProperty                     (RCA0)
L1 |   LogProperty                   (RCA0)
L1 |   PosClosedMul                  (RCA0)
L1 |   Def:exp                       (RCA0)
```

Theorem **ImageExpLn** (**RCA$_0$**)

```
L0 | ImageExpLn                     (RCA0)
L1 |   ax:R-order                    (ZF)
```

```
L1 |   Pos:ImgUnion                (FOL)
L1 |   Pos:LnBij                   (RCA0)
L1 |   Pos:LnSign                  (RCA0)
L1 |   Def:exp                     (RCA0)
L1 |   Def:ln                      (RCA0)
L1 |   Def:R-pos                   (RCA0)
```

Theorem **MVT**: two proofs, two bounds

Proof A (Barrow, §3; WKL$_0$).

```
L0 | MVT (Proof A)                (WKL0)
L1 |   EVT                        (WKL0)
L1 |   Pos:BarrowTangency         (RCA0)
L1 |   Def:chord                  (RCA0)
```

Proof B (Rolle-based, H–V; RCA$_0$).

```
L0 | MVT (Proof B)                (RCA0)
L1 |   Pos:HVRolle                (RCA0)
L1 |   Def:chord                  (RCA0)
```

The tag RCA$_0$ is the strength of Proof B; Proof A alone is WKL$_0$. Same split for Rolle, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity. Darboux and WeightedMVT are WKL$_0$ throughout.

Theorem **FFTC** (RCA$_0$)

```
L0 | FFTC                         (RCA0)
L1 |   BarrowArea                 (RCA0)
L2 |     Pos:IntMono              (RCA0)
L1 |   Pos:IntAdd                 (RCA0)
L1 |   Def:Area                   (RCA0)
L1 |   Def:Ck                     (RCA0)
```

12.3. Transitive closure

```
ImageExpLn                     (RCA0)
|-- ax:R-order                 (ZF)
|-- Pos:ImgUnion               (FOL)
|-- Pos:LnBij                  (RCA0)
|-- Pos:LnSign                 (RCA0)
|-- Def:exp                    (RCA0)
|-- Def:ln                     (RCA0)
`-- Def:R-pos                  (RCA0)

ExpProperty                    (RCA0)
|-- LogProperty                (RCA0)
|-- PosClosedMul               (RCA0)
`-- Def:exp                    (RCA0)

LogProperty                    (RCA0)
|-- PosClosedMul               (RCA0)
|-- Pos:IntLinear              (RCA0)
|-- Pos:IntAdd                 (RCA0)
|-- Pos:IntSubst               (RCA0)
|-- Def:ln                     (RCA0)
|-- ax:R-mul-comm              (ZF)
`-- ax:R-mul-inv               (ZF)

PosClosedMul                   (RCA0)
|-- ax:R-order                 (ZF)
|-- ax:R-mul-monotone-pos      (ZF)
`-- ax:R-mul-inv               (ZF)
```

12.4. Strength map

```
STRENGTH  |  ENTRIES
----------+-------------------------------------------------------------
FOL       |  Pos:ImgUnion
----------+-------------------------------------------------------------
RCA0      |  AXIOMS: all fourteen field axioms (ZF, no choice)
|  POSTULATES: IntLinear IntAdd IntMono IntSubst ContInt
|              Similarity ProdRule ChainRule InvCont
|              BarrowTangency LnBij LnSign
|              ContImgBdd Sigma01Choice SeqCont HVRolle YokoyamaInv
|  DEFS: L1 R-pos Ck strong-Ck Diff Area Chord Tangent
|        LocExt Increasing StrictInc StrictDec StrictMono ln exp
|  TOPOLOGY:  (none)
|  DIFF: Fermat Rolle MVT Const Monotone CauchyMVT
|        LHopital Taylor Convexity Inverse
|  INTEG: BarrowArea MVTInt FFTC SecondFTC IBP Subst
|  LOG/EXP: PosClosedMul LogProperty ExpProperty ImageExpLn
----------+-------------------------------------------------------------
WKL0      |  IVT, EVT, Darboux, WeightedMVT
|  (as printed; see EVT tag remark in \S\ref{sec:core-statements})
|  Pos:ContImgAttain, Pos:Pi01Sup
|  IVT (seq), MVT (seq)
|  Hahn-Banach, Weierstrass
----------+-------------------------------------------------------------
ACA0      |  separation (closed convex)
----------+-------------------------------------------------------------
ZF        |  ax:R-add-assoc ... ax:R-complete (all 14)
----------+-------------------------------------------------------------
ZFC       |  Banach-Alaoglu, Tychonoff, well-ordering
----------+-------------------------------------------------------------
```

12.5. Dependency depth

Axioms and postulates are depth 0; definitions depth 0 as primitive notation.

Depth	Entries (selection)
0	14 axioms, 20 postulates, 15 definitions
1	PosClosedMul, Fermat, BarrowArea
2	Rolle, MVT, LogProperty, ImageExpLn, MVTInt, FFTC
3	ExpProperty, Const, Monotone, SecondFTC
4	CauchyMVT, Convexity, IBP, Subst

Longest chain: $ax{:}R\text{-}order \to PosClosedMul \to LogProperty \to ExpProperty$ **(length 3). Diff/integ chains reach depth 4.**

12.6. Strength propagation

Result	Lower	Upper	Choice
IVT, EVT	WKL_0	WKL_0	none
Fermat	RCA_0	RCA_0	none
Rolle, MVT	RCA_0	RCA_0	none
Const, Monotone	RCA_0	RCA_0	none
CauchyMVT, LHopital, Taylor	RCA_0	RCA_0	none
Convexity	RCA_0	RCA_0	none
Inverse, MVTInt	RCA_0	RCA_0	none
Darboux, WeightedMVT	WKL_0	WKL_0	none
BarrowArea, FFTC	RCA_0	RCA_0	none

Bounds coincide: every RCA_0 tag has an RCA_0 proof in §11 (direct, or Hardin–Velleman, or Yokoyama); every WKL_0 tag has the classical proof. Part 1's geometric proofs of Rolle, MVT, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity are at most WKL_0 as written; the gap is disclosed in the EVT remark and does not affect any tag. No proof of Part 2 uses the Cut axiom.

13. Conclusion

Classical analysis developed in Barrow's geometry (Part 1) is classified in reverse mathematics (Part 2):

- 14 field axioms: ZF, no choice.
- Postulates: RCA_0-provable or first-order logical.
- RCA_0: Rolle, MVT, Fermat, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity, Inverse, FFTC, SecondFTC, IBP, Subst, ln/exp addition and image.
- WKL_0 as proved here: IVT, EVT, Darboux, WeightedMVT.
- No entry uses Cauchy completeness ($\mathsf{DC}_{\mathbb{R}}$-equiv) or Cut.

13.1. Wider hierarchy

Single-function MVT is RCA_0 [3]. IVT, EVT, and sequential IVT/MVT are WKL_0 [4, 5]. Hahn–Banach and Weierstrass are WKL_0 [4]. Separation is ACA_0 [15]. Banach–Alaoglu, Tychonoff, and well-ordering are ZFC [12, 13, 14]. Every Part 1 entry is choice-free.

13.2. Dependency analysis

The core chain has depth 3 with longest path $ax{:}R\text{-}order \to PosClosedMul \to LogProperty \to ExpProperty$; differentiation and integration chains reach depth 4.

The load-bearing nodes are ax:R-order, ax:R-mul-inv, Def:ln, Def:exp, Pos:LnBij, and PosClosedMul. The join of predecessor strengths never exceeds WKL_0 and never reaches $\mathsf{ZF}+\mathsf{DC}_{\mathbb{R}}$. Classical theorems are exactly those whose proofs need only area additivity Equation 5, rectangle comparison Equation 28, the differentiating postulates Equation 10–Equation 11, and (for the four WKL_0 entries) the standard WKL_0-strength principles. No choice principle is needed.

13.3. Formal verification

Lean 4 and Mathlib [10] contain many of these results; about 70.4% of Mathlib declarations depend on Classical.choice, and 13.1% of theorems have a choice-free statement with a choice-dependent proof [11]. The classification above is strictly stronger: it certifies the choice-free proofs.

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